

Muhammad Hanzla

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Data Scientist | ML Engineer | Data Analyst

Professional Summary

BS Software Engineering student, results-driven Data Scientist, ML Engineer and Data Analyst with end-to-end expertise in building ML pipelines from raw data wrangling to production deployment. Expert at translating complex datasets into actionable business insights through rigorous data storytelling and predictive modeling. Active Kaggle contributor with 15+ published projects spanning classification, computer vision, NLP, and time-series forecasting.

Education

National University of Modern Languages Islamabad

Bachelor's, Software Engineering

Currently in 4th Semester

September 2024 - June 2028

GPA: 3.83

Key Projects

Scene Classifier — ResNet-50

- Accuracy: 96% · Top-2: 99%
- Transfer learning on 25K images, Grad-CAM + Docker + FastAPI
- [Link to project](#)

Sentiment Analysis — NLP

- F1: 0.91 · Accuracy: 92%
- TF-IDF, BiLSTM, and fine-tuned DistilBERT on 50K IMDB reviews
- [Link to project](#)

Fraud Detection System

- AUC: 0.992 · AP: 0.88
- 284K transactions, 0.17% fraud · XGBoost + SHAP explainability
- [Link to project](#)

COVID-19 EDA Dashboard

- Global trends, wave analysis, 3 Plotly dashboards + Prophet forecast
- 50+ Visualizations · 90-Day Forecast
- [Link to project](#)

Research Interests

- **Explainable AI:** Interpretable machine learning models for transparent, trustworthy decision-making in real-world applications
- **Computer Vision:** Applying deep learning techniques to solve real-world perception and recognition problems
- **Natural Language Processing:** Large Language Models, transformer architectures applied NLP for low-resource languages

Skills

- **Data Sciences:** Python, Statistical Analysis, Data Analysis, Predictive Modeling, Data Mining, Time Series Analysis
- **Machine Learning:** Pytorch, Scikit-learn, Tensorflow, Deep Learning, Neural Networks, Natural Language Processing, Model Deployment, Computer Vision
- **Analytics Tools:** SQL, Power BI, Data Visualization, Tableau
- **Soft Skills:** Problem Solving, Critical Thinking, Attention to Detail, Self Learning, Team Collaboration, Communications
- **Languages:** Urdu, English

Achievements

- Achieved **Notebooks Expert** status on Kaggle, demonstrating high-quality contributions to the data science community
- Maintains a **CGPA of 3.83/4.0** in BS Software Engineering at NUML Islamabad, consistently among the top students in the program.